

Functional Documentation: Football Data Pipeline

Data Lake House Implementation on AWS for Advanced Analytics

Project developed for

Federación Mexicana de Fútbol (FMF)

Implemented by

Team 6 - Data Engineering, UNAM 2026.

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1. Document Objective

This document describes the system's functionalities from a sporting and strategic perspective for the user. It outlines what the system can do, who uses it, and what value it brings to the coaching staff of the Mexican national team, considering the upcoming matches at the 2026 World Cup.

2. User Roles

To ensure the correct flow of information and security, the following profiles are defined

Role	Responsibilities
Business Analyst	Queries metrics including wins, draws, losses, average goals, etc. Used to perform analyses and generate reports on rival teams at the 2026 World Cup.
Data Engineer (Support)	This team will focus on monitoring the extraction from API-Football and ensuring that AWS Glue processing runs correctly
Stakeholder / Executive (Coaching Staff)	Will review rival team statistics and history for decision-making and match planning for the 2026 World Cup through the dashboard

3. Supported Business Processes

3.1. Match Centralization

The system automatically consolidates the results of international matches played between August 2022 and May 2026. This data was extracted directly from API-Football and deposited in raw state into the Bronze layer in CSV format. This keeps a historical record of the data and eliminates the need to manually search and record information.

1	fecha	local	visitante	goles_local	goles_visitant	torneo	
2	2022-11-22T1	Mexico	Poland	0	0	World Cup	
3	2022-11-26T1	Argentina	Mexico	2	0	World Cup	
4	2022-11-30T1	Saudi Arabia	Mexico	1	2	World Cup	
5	2023-03-24T0	Suriname	Mexico	0	2	CONCACAF Nations League	
6	2023-03-27T0	Mexico	Jamaica	2	2	CONCACAF Nations League	
7	2022-09-01T0	Mexico	Paraguay	0	1	Friendlies	
8	2022-09-25T0	Mexico	Peru	1	0	Friendlies	
9	2022-09-28T0	Mexico	Colombia	2	3	Friendlies	
10	2022-11-09T2	Mexico	Iraq	4	0	Friendlies	
11	2022-11-16T1	Mexico	Sweden	1	2	Friendlies	
12	2023-06-16T0	USA	Mexico	3	0	CONCACAF Nations League	
13	2023-06-18T2	Panama	Mexico	0	1	CONCACAF Nations League	
14	2023-04-20T0	USA	Mexico	1	1	Friendlies	
15	2023-06-26T0	Mexico	Honduras	4	0	CONCACAF Gold Cup	
16	2023-06-30T0	Haiti	Mexico	1	3	CONCACAF Gold Cup	
17	2023-07-03T0	Mexico	Qatar	0	1	CONCACAF Gold Cup	
18	2023-06-08T0	Mexico	Guatemala	2	0	Friendlies	
19	2023-06-11T0	Mexico	Cameroon	2	2	Friendlies	
20	2023-07-09T0	Mexico	Costa Rica	2	0	CONCACAF Gold Cup	
21	2023-07-13T0	Jamaica	Mexico	0	3	CONCACAF Gold Cup	
22	2023-07-16T2	Mexico	Panama	1	0	CONCACAF Gold Cup	
23	2023-09-10T0	Mexico	Australia	2	2	Friendlies	
24	2023-09-12T2	Mexico	Uzbekistan	3	3	Friendlies	
25	2023-10-15T0	Mexico	Ghana	2	0	Friendlies	
26	2023-10-18T0	Mexico	Germany	2	2	Friendlies	

3.2. Automatic Data Cleansing

Once the raw data is available, an automated process runs that detects and corrects inconsistencies automatically. Quality rules are also applied, such as the elimination of null values, standardization of team names, and validation of goal values within a range of 0 to 20.

Finally, all results are stored in Parquet format, partitioned by year, ensuring that subsequent analysis is based on a single version.

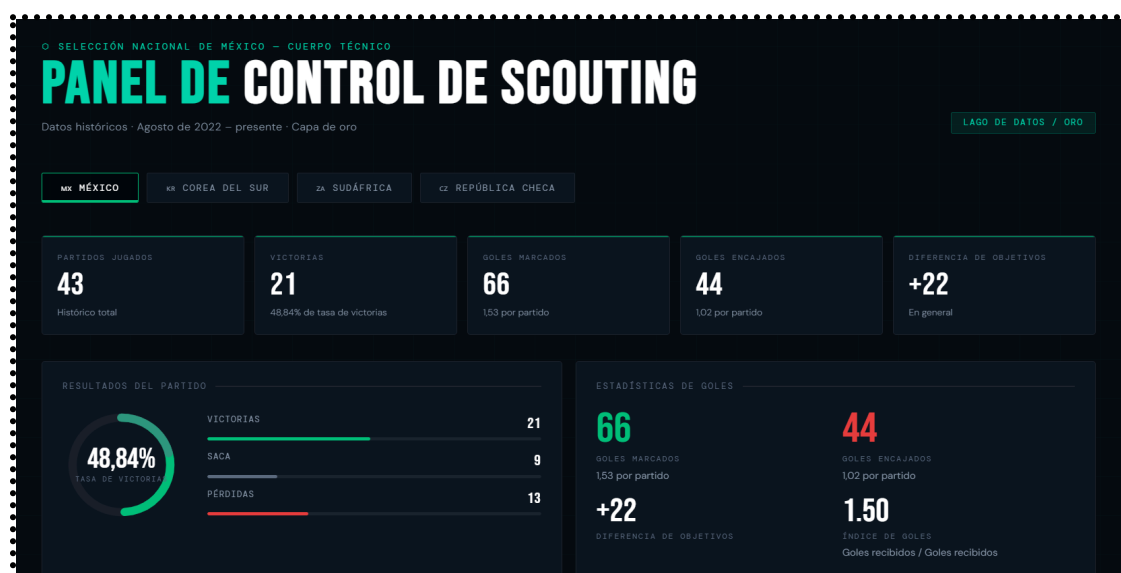


#	fecha	local	visitante	goles_local	goles_visitante	torneo	partition_0
1	2022-09-01 01:00:00.000	MEXICO	PARAGUAY	0	1	Friendlies	mexico
2	2022-09-25 01:00:00.000	MEXICO	PERU	1	0	Friendlies	mexico
3	2022-09-28 02:00:00.000	MEXICO	COLOMBIA	2	3	Friendlies	mexico
4	2022-11-09 20:00:00.000	MEXICO	IRAQ	4	0	Friendlies	mexico
5	2022-11-16 19:30:00.000	MEXICO	SWEDEN	1	2	Friendlies	mexico
6	2022-11-22 16:00:00.000	MEXICO	POLAND	0	0	World Cup	mexico
7	2022-11-26 19:00:00.000	ARGENTINA	MEXICO	2	0	World Cup	mexico
8	2022-11-30 19:00:00.000	SAUDI ARABIA	MEXICO	1	2	World Cup	mexico

3.3. Historical Reporting

This system allows the coaching staff of the Mexican football federation to perform comparisons evaluating Mexico's potential rivals, as well as the year-over-year evolution from 2022 through 2026.

Thanks to the historical accumulation of data in the final layer, it is possible to run queries using Amazon Athena to extract strategic metrics, such as the win percentage of rival teams, goals scored or conceded, and overall performance by tournament type. With this data, it is possible to determine whether a team tends to be more offensive or defensive, allowing the coaching staff to plan the most appropriate lineup for the match.



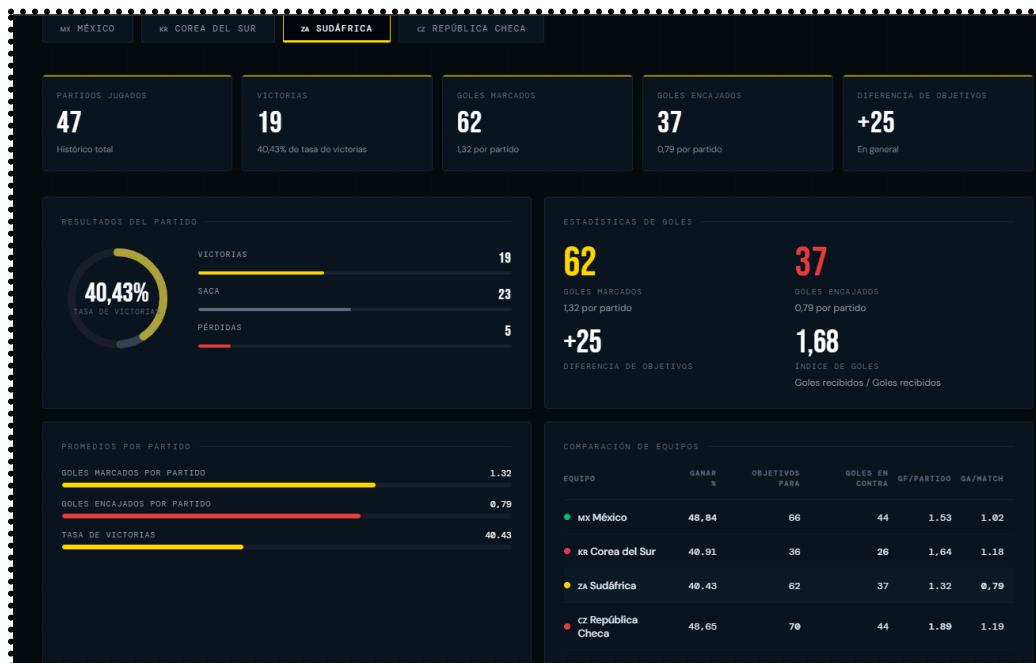
4. Use Cases

Caso 1:

The tactical analyst wants to prepare for the upcoming June 11 match against South Africa.

Result:

The tactical analyst queries Amazon Athena for South Africa's history since 2022. The system returns the KPIs available in the Gold Layer: win, draw, and loss percentages, average goals scored and conceded per match, and differentiated performance by tournament type (friendlies vs. official FIFA matches). Based on these metrics, the coaching staff can identify whether South Africa tends to be an offensive or defensive team and plan the most suitable lineup for the match.



Caso 2:

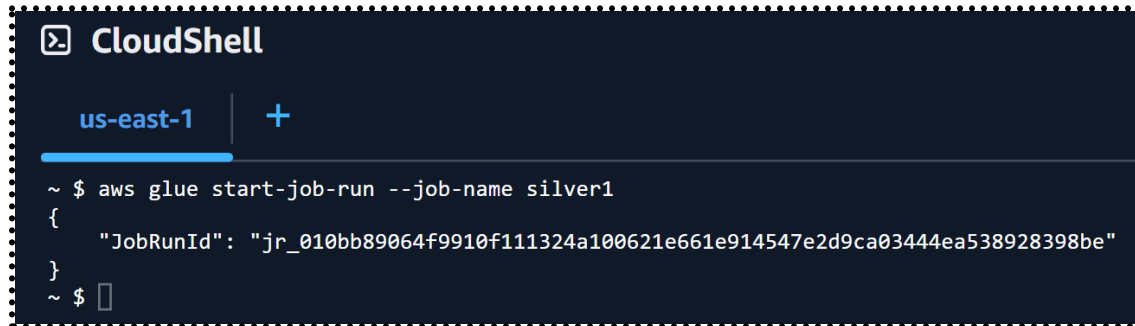
An error is detected in the original source.

Result:

The system allows performing a *backfill* or reprocessing to correct historical data in full. The Data Engineer corrects the affected files in the Bronze Layer and re-runs the Glue Job silver1 using the command

aws glue start-job-run --job-name silver1.

The pipeline overwrites the corrected Parquet files in the Silver Layer, the Glue Crawler automatically updates the catalog, and the data becomes available in Athena without any additional intervention.



```
CloudShell

us-east-1 | +

~ $ aws glue start-job-run --job-name silver1
{
  "JobRunId": "jr_010bb89064f9910f111324a100621e661e914547e2d9ca03444ea538928398be"
}
~ $
```

5. Business Rules

1. Only matches with goal values between 0 and 20 per team are processed. Records outside this range are automatically discarded by the pipeline as anomalous data.
2. The names of national teams (home_team and away_team) must always be stored in uppercase and without leading or trailing spaces to ensure consistency in searches and groupings.
3. The date field must be valid and convertible to Timestamp type. Data is mandatorily partitioned by team and by year within S3 to ensure query efficiency in Amazon Athena.

6. User Glossary

Bronze Layer (Raw Data)

First layer of the Data Lake where data is stored exactly as it arrives from API-Football, in CSV format, with no transformation applied. It is the pure backup of the historical record.

Silver Layer (Clean Data)

Second layer where data has already been cleansed, typed, and standardized by the Glue Job. Stored in Parquet format, partitioned by team and year. It is the system's single source of truth.

Gold Layer (KPIs and Metrics)

Final consumption layer containing aggregated strategic indicators (wins, draws, losses, goals, and win rate percentage per team). This is the layer directly queried by the coaching staff through Amazon Athena.

Away Team

National team playing outside its home country or official venue during a match.

Draw

Match result in which both teams finish with the same number of goals.

Fixture

Official football match between two national teams.

Goals Against

Total number of goals conceded by a team.

Goals For

Total number of goals scored by a team.

Group Stage

Initial phase of a tournament where teams compete within groups before advancing to the knockout stage.

Historical Matches

Previously played matches used for statistical and analytical purposes.

Knockout Stage

Tournament phase where the losing team is immediately eliminated.

Losses

Total number of matches lost by a team.

Per Match Average

Average statistical value calculated per game, such as goals scored or conceded.

National Team

Official football team representing a country in international competitions.

Opponent

Rival team facing another team during a football match.

Points

Scoring system commonly used in football tournaments:

- Win = 3 points
- Draw = 1 point
- Loss = 0 points

Rival

Competing national team analyzed within the project.

Scoreline

Final result of a football match.

Example: Mexico 2 - 1 South Korea.

Tournament

Official football competition involving national teams, such as the FIFA World Cup or Gold Cup.

Victories

Total number of matches won by a team.

Win Percentage

Percentage of victories relative to total matches played.

Wins

Matches in which a team scored more goals than its opponent.